SkinSmith: A Constraint-Aware Generative Agent for Game Weapon Skin Design

Turn a single keyword (e.g. "dragon") into a mapped, seam-compliant CS2 weapon texture — with three blocking human checkpoints.

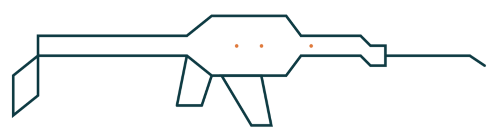
1 · Background & Problem Why image generation alone is insufficient for mapped textures

- Modern text-to-image models (SD-Turbo, Gemini-Image, …) can produce stunning rectangular art — but a game skin is not a rectangle.
- **UV fragmentation:** a 3D surface is unfolded, scaled, and repacked into an atlas, shredding any continuous pattern.
- **View inconsistency:** art that looks great on the left side may collapse from the top.
- **Export constraints:** hard rules on resolution, drawable area, seams, and palette.
- Pretty source image ≠ deployable weapon texture.

Source art · dragon master artwork generated after CP2 and evaluated at CP3

2 · Aims *What this project is — and is not*

- Build a constraint-aware generative Agent: theme keyword → mapped CS2 texture candidate.
- Apply hard feasibility rules (seam budget and fixed multi-view checks) before export.
- Test adapter-level transfer on M4A4 without claiming equivalent acceptance.
- Academic prototype — not affiliated with Valve; not published to Steam Workshop.



3 · Three Blocking Human Checkpoints *The agent never exports without approval*

3 blocking human checkpoints — agent never exports without approval

CP1 · Theme pack
Confirm the visual world: concept, palette, motifs, mood, forbidden content.

CP2 · Artistic direction
Pick 1 of 3–4 text-prompted style directions. Model recommends, user decides.

CP3 · Final mapped artwork
See source + 3 mapped views (left / right / top) before approving any export.

4 · Three Principles *What we learned — and what we recommend*

1. Bind the source image to the mapping view.
   Never approve art on a flat canvas — the human must see how it sits on the gun.
2. Feasibility ≠ preference.
   Treat export readiness as a hard gate, not as another weighted feature in the score.
3. Corrections have a budget and a rollback rule.
   Small "reasonable" edits can erase detail. Roll back unless the score genuinely improves.

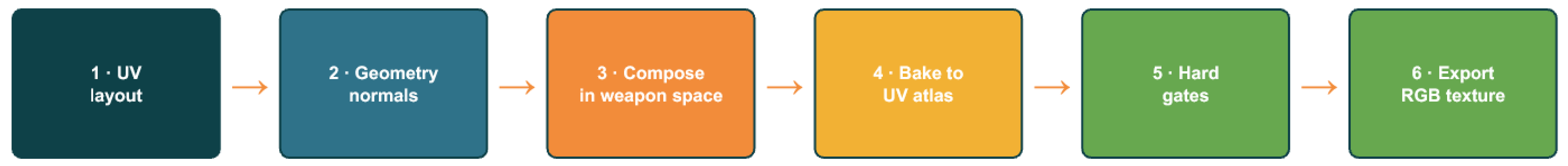
5 · What We Deliver *Concrete outputs and how to reproduce*

- Pipeline: Streamlit + CLI for the CP1 → CP2 → CP3 → export workflow.
- Models: SD-Turbo (local), Gemini-3.1 Flash Image (API), and Flash-Lite planning.
- Evidence: machine-readable JSON/CSV logs and 100 automated tests.
- Output: 2048 × 2048 RGB PNG/TGA textures baked to the AK-47 UV atlas.

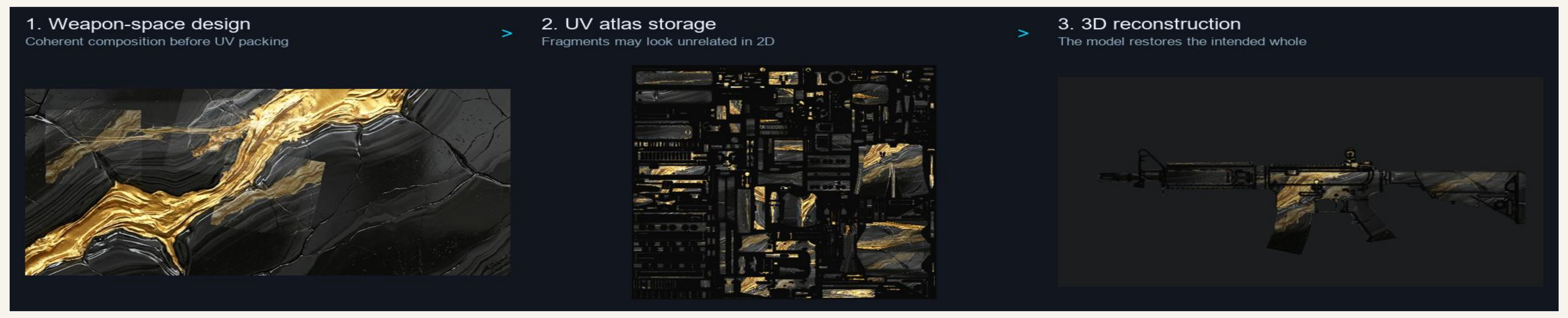
6 · Route B: Weapon-Space Composition *Compose in 3D, then bake to UV atlas*

- Route A (tileable baseline): compose a crop-tolerant 2D pattern, then map it through the asset UV.
- Route B (this work): compose continuously in weapon space, then use geometry-normal guidance to bake back to the UV atlas.

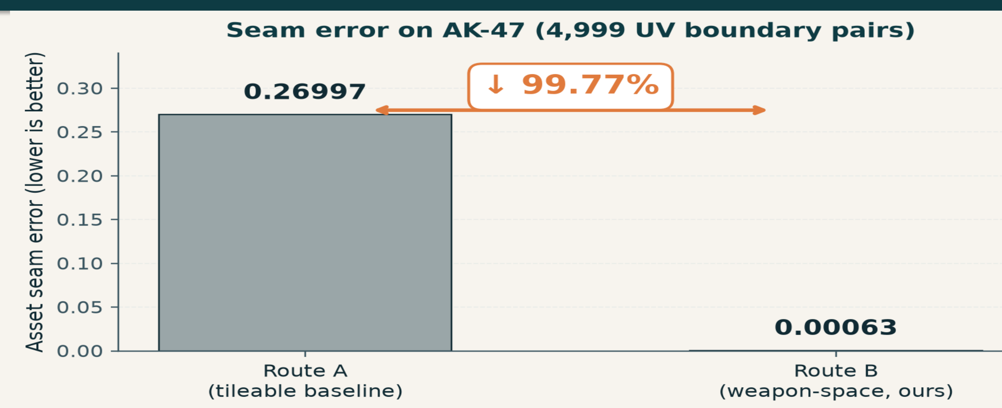
Substantially reduces true mapped seam error while preserving multi-view content.



Geometry normals guide baking; the exported asset is base-colour only.



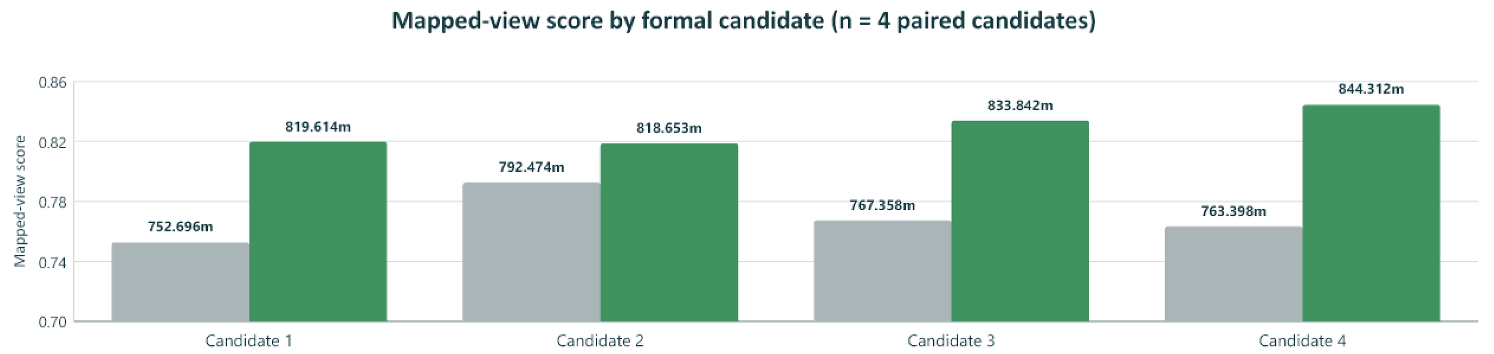
7 · True Seam Error on AK-47 *Geometric discontinuity of 4,999 3D-adjacent UV boundary pairs*



8 · Hard Constraints the Agent Enforces *Feasibility is not preference — it is a gate*

Seam budget ≤ 0.01 / edge	**Palette lock** user-locked set
No repeat anti-periodic UV	**Multi-view OK** 3 / 3 fixed views

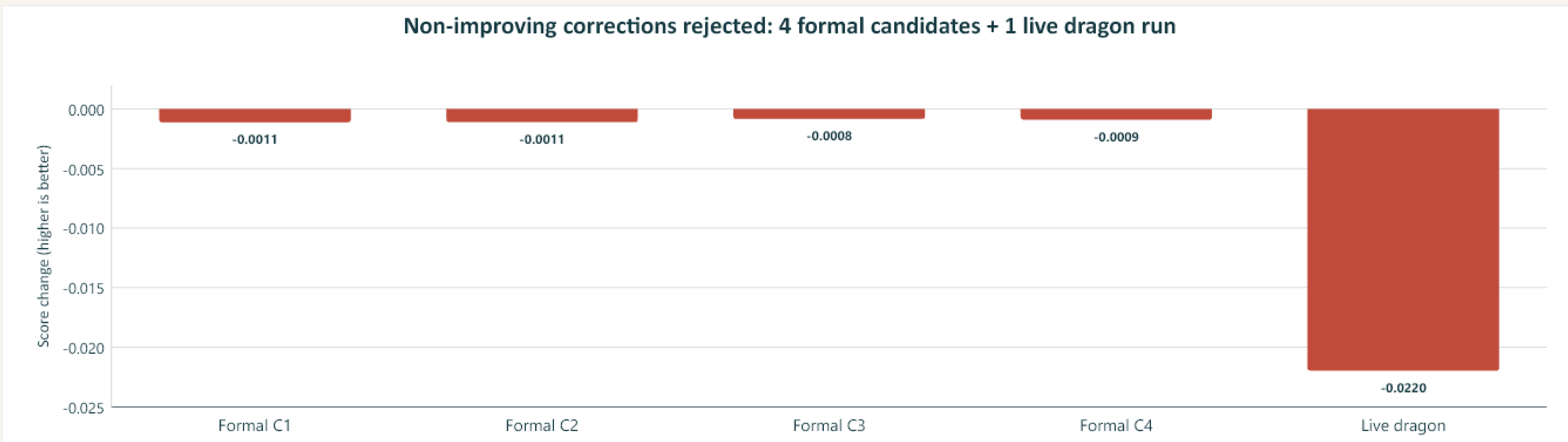
9 · Paired Multi-View Result · 4 / 4 Route B improves mapped-view score for all four formal candidates



10 · Selection & Self-Correction *Feasibility > preference · Route C safely rejected*

Weighted score (early)
seam = 0.0166
✗ FAILS export test — small visual gain hid a export-gate failure.

Constraint-first + Pareto (current)
seam = 0.0055
✓ EXPORTABLE — passes texture export gates.



11 · Key Results on AK-47 *Route B vs Route A — head to head*

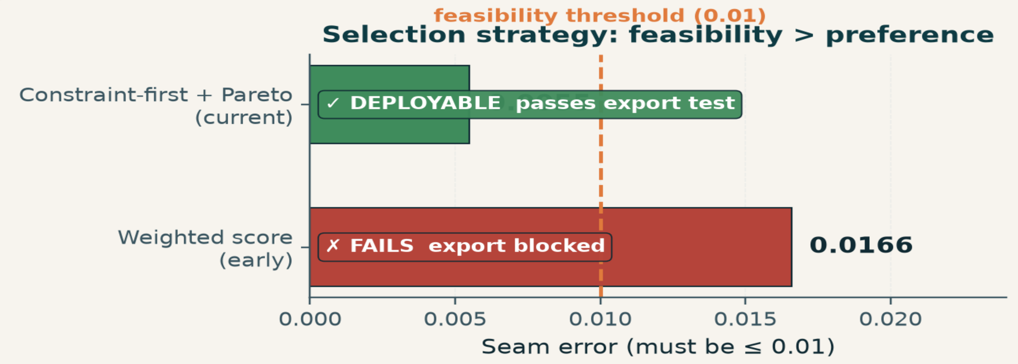
ASSET SEAM ERROR
▼ 99.77%
0.00063 vs 0.26997
(Route B vs Route A)

MULTI-VIEW SCORE
▲ +0.060
0.829 vs 0.769
4 / 4 candidates improve

12 · Real Run: dragon → Treasured Relic *The end-to-end run that produced the final export*

- Selected: artwork_04 (1 of 4)
- Seam error: 0.00078 (< 0.01)
- Multi-view: 0.8166 (left / right / top)
- Export: 2048 × 2048 RGB PNG/TGA

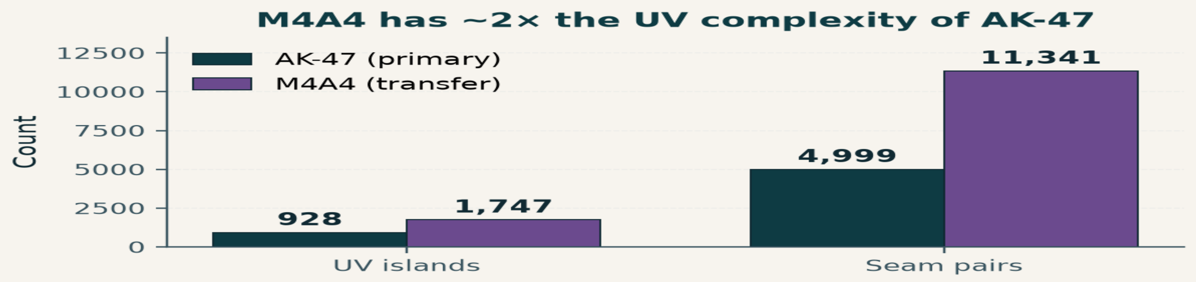
13 · Selection Strategy: Feasibility > Preference Constraint-first + Pareto selects a feasible export candidate



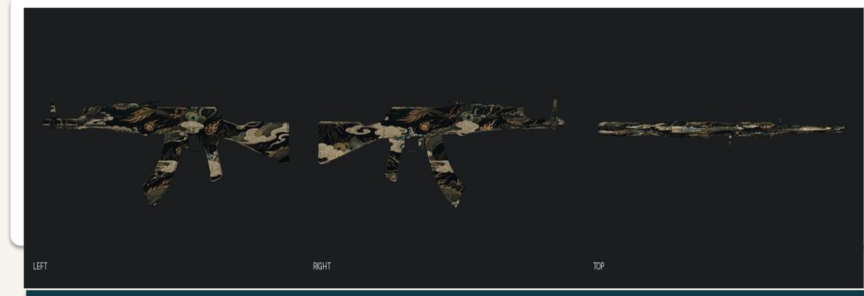
14 · Adapter Transfer: AK-47 → M4A4 *Agent contract is reusable · asset params must be re-calibrated*

REUSABLE (purple)
- Agent state machine
- Checkpoint contract
- Generation back-end
- Eval & logging

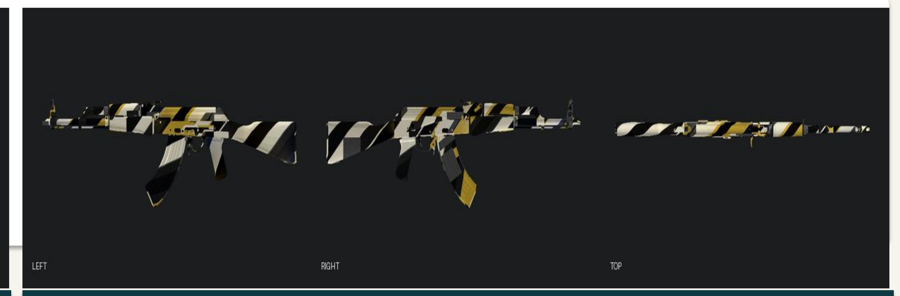
MUST RE-CALIBRATE (amber)
- OBJ / UV binding
- Seam map
- Camera poses
- Mask, edge width, thresholds



15 · Visual Gallery · Route B Outputs LEFT / RIGHT / TOP software renders of two mapped candidates



Treasured Relic (artwork_04)



Hazard Sigil (alt. style)

16 · Conclusion *What we deliver*

- Deliver: mapped, seam-compliant CS2 texture candidates with three human checkpoints.
- Validate: 100 automated tests; code on GitHub; AK-47 primary, M4A4 adapter transfer.
- Safety result: four formal corrections and the live dragon correction were rejected as non-improving.

17 · Limitations & Future Work *Honest about what this prototype does not yet do*

- Weapons: AK-47 complete; M4A4 is adapter-level transfer evidence only.
- Materials: base-colour only; no normal, roughness, metallic, height, or full PBR output.
- Deployment: Streamlit is the demo client; no automated Steam Workshop publication.
- Demo: python -m streamlit run streamlit_app.py

Code & Evidence
github.com/DID-FIELD/SDSC6002-SkinSmith
100 automated tests · machine-readable logs · versioned exports

Route B

SkinSmith · Constraint-Aware Generative Agent for Game Weapon Skin Design

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